SOFTWARE REQUIREMENTS SPECIFICATION

Advanced Hospital Management System

Comprehensive HMS Platform — Patient Care, Clinical, Operational & Financial Management

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1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for the Advanced Hospital Management System (HMS) — a centralized, multi-department platform that digitizes and automates the clinical, administrative, and financial operations of a hospital or clinic network. It is intended for use by the project owner, business analysts, UI/UX designers, backend and frontend engineers, QA engineers, and DevOps personnel involved in building and maintaining the system.

1.2 Scope

The HMS will provide a single source of truth for patient data, clinical workflows, laboratory and pharmacy operations, billing and finance, human resources, inventory, and hospital-wide reporting. The system is designed to be modular and multi-hospital ready, allowing a single deployment to eventually serve multiple hospital branches or a hospital group, while the MVP targets a single medium-sized hospital or clinic. The platform will expose a REST API layer so that mobile applications and government reporting systems can integrate with it in later phases.

1.3 Intended Audience

Audience	Interest
Product Owner / Hospital Administration	Validating that the system covers real operational workflows
Development Team	Technical implementation of modules, APIs, and data model
QA / Test Team	Deriving test cases and acceptance criteria
UI/UX Designers	Screen flows, role-based dashboards
Compliance / Legal	Data protection, medical record retention, audit requirements

1.4 Definitions, Acronyms and Abbreviations

Term	Meaning
HMS	Hospital Management System
EMR / EHR	Electronic Medical Record / Electronic Health Record
OPD / IPD	Out-Patient Department / In-Patient Department
OT	Operation Theater
ER	Emergency Room / Emergency Department
RBAC	Role-Based Access Control
PACS	Picture Archiving and Communication System
HL7 / FHIR	Health Level 7 / Fast Healthcare Interoperability Resources — healthcare data exchange standards
SLA	Service Level Agreement
MVP	Minimum Viable Product

1.5 References

This SRS expands on the original hospital management system requirements outline supplied by the project owner, restructured against standard SRS conventions (IEEE 830 style) and enriched with modules commonly required in production-grade hospital systems used in South Asian healthcare markets, including Bangladesh-specific considerations such as Bengali/English bilingual support and mobile-banking payment methods.

2. Overall Description

2.1 Product Perspective

The HMS is a new, standalone, web-based (with future mobile companion apps) system. It follows a modular, service-oriented architecture so that individual modules (Laboratory, Pharmacy, Billing, HRM, etc.) can be developed, tested, scaled, and deployed with a degree of independence, while sharing a common patient, user, and permission model.

2.2 Product Functions (Summary)

At a high level the system allows front-desk staff to register and schedule patients, doctors to record consultations and prescriptions, lab and radiology staff to process diagnostic orders, pharmacists to dispense medicine against verified prescriptions, accountants to generate and reconcile bills, HR staff to manage employees, and administrators to configure the hospital, monitor KPIs, and generate reports — all under a single audited, role-restricted platform.

2.3 User Classes and Characteristics

See Section 3 for the full role and permission matrix. In summary, the system supports Super Admin, Hospital Admin, Doctor, Nurse, Receptionist, Lab Technician, Radiologist, Pharmacist, HR Manager, Accountant, Store/Inventory Manager, Ward Boy/Support Staff, and Patient (self-service portal) roles.

2.4 Operating Environment

- Web application accessible via modern browsers (Chrome, Edge, Firefox, Safari) on desktop and tablet.
- Backend deployed on Linux-based cloud infrastructure (containerized), with horizontal scalability.
- Companion mobile apps (future phase) for Android and iOS for patients and doctors.
- Peripheral hardware support: barcode/QR scanners, receipt/label printers, biometric attendance devices, RFID readers for patient wristbands, vital-sign monitor integration (future).

2.5 Design and Implementation Constraints

- Must comply with local data protection expectations for medical records (retention, access logging) and should be architected to support HIPAA/GDPR-style controls for future international deployment.
- Bilingual UI (English and Bengali) is required from MVP given the target market.
- Must support offline-tolerant behavior at least for critical desk operations (e.g., queue token printing) during short network interruptions.

2.6 Assumptions

The hospital has continuous internet connectivity at each facility, staff will receive training on the new system, and a data-migration exercise will move any existing patient/employee records into the new platform prior to go-live.

3. User Roles & Permission Matrix

The system uses Role-Based Access Control (RBAC) with support for custom roles composed from granular permissions (see Module 22). The table below expands the original role list and adds roles required for a complete hospital workflow (Radiologist, Store/Inventory Manager, OT Manager, Ward Boy/Support Staff).

Role	Core Responsibilities	Key Permissions
Super Admin	Owns the entire platform across all hospital branches	Hospital onboarding, global settings, subscription/billing of the SaaS itself, system-wide audit logs, user impersonation for support
Hospital Admin	Runs a single hospital/branch	Department setup, staff onboarding, patient oversight, billing oversight, inventory oversight, report access
Doctor	Clinical consultation and treatment	View assigned appointments, patient history, write e-prescriptions, order lab/radiology tests, recommend admission, view own performance reports
Nurse	Ward and patient care	Record vitals, administer/update medication, bed management, nursing notes, IPD monitoring
Receptionist / Front Desk	Patient-facing registration and scheduling	Register patients, book/reschedule appointments, generate tokens, collect payments at desk, patient search
Lab Technician	Diagnostic sample processing	Receive test requests, log sample collection, enter results, upload/print reports
Radiologist / Imaging Technician **ADDED**	Imaging studies	Receive imaging orders, upload images/reports, mark studies complete
Pharmacist	Medicine dispensing	Verify prescriptions, dispense medicine, update stock, flag drug interactions
HR Manager	Workforce administration	Employee records, payroll, leave, attendance, recruitment, shift scheduling
Accountant / Finance Officer	Billing and financial reporting	Billing, expenses, income, ledgers, P&L, reconciliation
Store / Inventory Manager **ADDED**	Non-pharmacy stock control	Purchase orders, stock transfers, minimum-stock and expiry alerts, vendor management
OT Manager / Scheduler **ADDED**	Operation theater coordination	OT scheduling, equipment/staff allocation, OT billing capture
Ward Boy / Support Staff **ADDED**	Patient transport, housekeeping coordination	Bed status updates (cleaning/maintenance), task acknowledgment
Patient (Self-Service Portal)	Own care and administration	Book appointments, view own EMR summary, reports, prescriptions, bills, make payments, telemedicine (future)

> All roles are configurable: Hospital Admins can create custom roles by combining granular permissions rather than being limited to the roles listed above (see Module 22 — Role & Permission Management).

4. Functional Requirements — Modules

Each module below lists its purpose, detailed features/fields, and representative functional requirement IDs in the form `FR-[Module].[n]` to support traceability into test cases. Modules marked NEW were not present in the original outline and are proposed as necessary additions for a production-grade HMS.

Module 1 — Dashboard

Role-aware landing page; widget set differs by role (e.g., a Doctor sees "My Appointments Today", an Accountant sees "Pending Collections").

Widgets

- Today's Patients, Today's Revenue, Admissions, Discharges
- Doctors Available, Bed Occupancy (with ward-wise breakdown), Appointment Queue
- Pending Bills, Pending Lab Tests, Low-Stock Alerts NEW
- Critical/Emergency Alerts feed NEW
- Staff on Leave Today NEW

ID	Requirement
FR-DSH.1	The system shall render a role-specific dashboard within 2 seconds of login.
FR-DSH.2	The system shall allow widgets to be rearranged and hidden per user preference.
FR-DSH.3	Dashboard KPIs shall refresh at least every 60 seconds or on manual refresh.

Module 2 — Patient Management

Patient Registration Fields

- Patient ID (auto-generated, unique, human-readable e.g. HMS-2026-000123), Name, Photo, Gender, DOB, Blood Group, Mobile, Email, Address, Emergency Contact, Known Allergies
- National ID / Birth Registration Number NEW, Marital Status, Occupation, Referring Doctor/Source NEW
- Guardian details for minors NEW, preferred language (English/Bengali) NEW

Functions

- Search Patient (by name, phone, ID, NID), Merge Duplicate Patients, Patient Timeline, Visit History
- Patient consent capture and e-signature for procedures NEW
- QR/barcode wristband generation for admitted patients NEW
- Family linking (link dependents under one guardian account) NEW

ID	Requirement
FR-PAT.1	The system shall prevent duplicate patient records by fuzzy-matching name, DOB and phone at registration, prompting staff to confirm or merge.
FR-PAT.2	The system shall maintain a complete chronological visit timeline per patient across OPD, IPD, ER and lab visits.
FR-PAT.3	The system shall generate a unique, scannable patient ID at registration.

Module 3 — Appointment Management

- Doctor Schedule (slot-based, configurable slot length per doctor/department)
- Online Booking, Walk-in Registration, Queue Management, Token System (digital display board support NEW)
- SMS Reminder, Email Reminder, WhatsApp Reminder NEW
- Appointment Reschedule, Cancellation, No-show tracking NEW
- Doctor leave/holiday blocking that automatically closes booking slots NEW

ID	Requirement
FR-APT.1	The system shall prevent double-booking of a doctor's time slot.
FR-APT.2	The system shall send automated SMS/email reminders 24 hours and 1 hour before an appointment.
FR-APT.3	The system shall display a live queue/token status on a public-facing screen and the patient portal.

Module 4 — Electronic Medical Record (EMR)

- Diagnoses (ICD-10 coded NEW), Symptoms, Prescriptions, Lab Results, Radiology, Allergies, Vaccinations/Immunization schedule, Procedures, Clinical Notes, Attachments
- Vital signs trend charts (BP, pulse, temperature, SpO2, weight/height, BMI) NEW
- Growth charts for pediatric patients NEW
- Structured templates per specialty (cardiology, gynecology, pediatrics, etc.) NEW
- Version history / amendment trail for clinical notes (immutable audit trail) NEW

ID	Requirement
FR-EMR.1	The system shall record every clinical note with author, timestamp, and an immutable version history; edits create a new version rather than overwriting.
FR-EMR.2	The system shall allow a doctor to view a consolidated timeline of a patient's diagnoses, prescriptions and lab results from any prior visit within the hospital.
FR-EMR.3	The system shall support attaching scanned documents and images to a visit record (max configurable file size).

Module 5 — Doctor Management

- Qualifications, Departments, Schedule, Consultation Fee, Availability
- Doctor rating/feedback from patients NEW
- Revenue-share / commission configuration per doctor NEW
- Digital signature for e-prescriptions NEW

Module 6 — Laboratory Management

- Lab Categories: Hematology, Biochemistry, Microbiology, Pathology, Serology NEW
- Test Catalog with reference/normal ranges by age & gender NEW, Test Requests, Sample Collection, Barcode, Sample Tracking, Test Processing, Result Entry, Approval, Report Generation
- Critical value auto-flagging and alert to ordering doctor NEW
- Lab equipment/analyzer interfacing (HL7/ASTM) for auto result import (future) NEW

ID	Requirement
FR-LAB.1	The system shall generate a barcoded sample label at the point of collection linked to the test request.
FR-LAB.2	The system shall automatically flag results outside critical reference ranges and notify the ordering doctor.
FR-LAB.3	Lab reports shall require technologist entry and a separate pathologist/senior approval step before release.

Module 7 — Radiology

- X-Ray, CT Scan, MRI, Ultrasound, ECG, Echo
- Imaging Request, Report Upload, Image Viewer, PACS Integration (Future)
- Radiologist worklist with priority flags (routine/urgent/stat) NEW

Module 8 — Pharmacy Management

- Medicine Database (generic name, brand, strength, form)
- Inventory: Purchase, Sales, Stock, Expiry, Batch Number, Barcode
- Prescription Dispensing with automatic dose/interaction check against patient allergies NEW
- Supplier Management
- Controlled/narcotic drug register with extra approval workflow NEW
- Partial dispensing and refill tracking NEW

Point of Sale (POS) Counter NEW

- Fast barcode/search-based cart for both prescription and over-the-counter (walk-in, no-prescription) sales
- Auto-pulls active e-prescriptions into the cart for a scanned/searched patient, with manual item add for OTC purchases
- Batch/expiry auto-selection using First-Expiry-First-Out (FEFO) logic, with manual batch override
- Line-item and bill-level discounts, configurable per staff permission level
- Split payment across Cash, Card, and Mobile Banking (bKash/Nagad/Rocket) within a single sale
- Hold/resume sale (park a cart to serve another customer, then resume)
- Instant thermal-printer receipt and reprint of past receipts
- Returns/exchange against original receipt with automatic stock reversal
- End-of-shift cash drawer reconciliation (expected vs. counted cash, over/short report)
- Real-time stock decrement on sale, feeding the same inventory ledger used by Module 9

ID	Requirement
FR-PHM.1	The system shall block dispensing of a medicine the patient is recorded as allergic to unless the pharmacist provides an override reason.
FR-PHM.2	The system shall prevent sale of expired or recalled batches at the point of sale.
FR-PHM.3	The POS counter shall complete a standard sale transaction (search/scan to receipt) in under 3 clicks/scans per item. NEW
FR-PHM.4	The POS shall support splitting a single sale's payment across more than one payment method and record each method's amount separately for reconciliation. NEW
FR-PHM.5	The POS shall generate a shift-close report showing total sales, returns, discounts, and expected vs. counted cash for each cashier. NEW

Module 9 — Inventory Management

- Inventory categories: Medical Equipment, Consumables, Medicines, Surgical Items, Linen/Housekeeping supplies NEW
- Purchase Orders, Stock Transfer (inter-department/inter-branch), Minimum Stock Alert, Expiry Alert
- Asset tagging and maintenance schedule for equipment NEW
- Vendor performance tracking NEW

Module 10 — Billing

- Invoices, Payments, Refunds, Discounts/waivers with approval workflow NEW
- Payment Methods: Cash, Card, Mobile Banking (bKash/Nagad/Rocket NEW), Bank Transfer
- Package/bundle billing (e.g., delivery package, surgery package) NEW
- Deposit/advance payment tracking for admitted patients NEW

Module 11 — Admission Management (IPD)

- Admission, Transfer, Bed Allocation, Ward Allocation, ICU, Discharge
- Discharge summary generation with follow-up instructions NEW
- Estimated cost calculator at admission based on package/ward rate NEW

Module 12 — Bed Management

- Bed Status: Available, Reserved, Occupied, Cleaning, Maintenance
- Ward Management with visual floor-plan bed map NEW

Module 13 — Operation Theater (OT)

- Schedule, Doctors, Equipment, Operation Notes, Billing
- Pre-op checklist and consent capture NEW, post-op recovery tracking NEW

Module 14 — Emergency Department (ER)

- Emergency Registration, Triage (with standard triage-level color coding NEW), Doctor Assignment, Admission, Critical Alerts

Module 15 — Ambulance Management

- Driver, Vehicle, GPS (Future), Trip Logs, Fuel, Maintenance
- Dispatch request from ER/reception with live status NEW

Module 16 — HRM

- Employee, Departments, Designation, Attendance, Leave, Payroll, Recruitment, Performance, Shift Management, Employee Documents
- Biometric/RFID attendance integration NEW, overtime and duty-roster management NEW

Module 17 — CRM

- Patient Follow-up, Campaigns, Feedback, Complaints, SMS, Email, Patient Loyalty, Marketing, Call Logs
- Automated post-discharge satisfaction survey NEW

Module 18 — Finance

- Income, Expenses, Accounts, Budgets, Profit & Loss, Balance Sheet, Cash Flow
- Department-wise cost/revenue center reporting NEW

Module 19 — Reports

- Patient, Doctor, Revenue, Lab, Inventory, HR, Finance reports
- Custom report builder with export to Excel/PDF NEW, scheduled email reports NEW

Module 20 — Notifications

- SMS, Email, Push Notifications, Internal Notifications
- WhatsApp Business API channel NEW

Module 21 — Document Management

- Upload, PDF, Scans, Consent Forms, Medical Certificates
- Document expiry tracking (e.g., staff licenses) NEW

Module 22 — Role & Permission Management (RBAC)

- Custom Roles, Permissions, Approval Workflow
- Field-level and record-level access restrictions (e.g., psychiatric notes visible only to treating doctor) NEW

Module 23 — Audit Logs

- Every action stored: Login, Update, Delete, Payment, Prescription, Reports
- Tamper-evident log (write-once) with export for compliance review NEW

New Modules Recommended for a Complete HMS NEW

Module 24 — Blood Bank Management NEW

- Donor registry, blood group/component inventory, cross-matching, issue/return tracking, expiry alerts for blood units, donor eligibility screening.

Module 25 — Consent, Privacy & Compliance Management NEW

- Digital consent forms for procedures/data sharing, configurable data-retention policies, right-to-access/erasure request handling, compliance dashboards.

Module 26 — Telemedicine / Video Consultation NEW

- Video/voice consultation, e-prescription issuance post-consult, waiting-room queue, session recording (with consent), integrated payment before session.
- Recommended as an early Phase-2 module rather than far-future, given post-pandemic patient expectations.

Module 27 — Patient & Doctor Mobile Apps NEW

- Patient app: booking, EMR summary, reports, bill pay, reminders, telemedicine.
- Doctor app: schedule, patient list, quick prescription, push alerts for critical lab values.

Module 28 — System Configuration & Localization NEW

- Hospital profile/branding, department/ward setup, tax and currency configuration, printable-document templates, bilingual (English/Bengali) label management, working-hours and holiday calendar.

Module 29 — Backup, Data Export & Disaster Recovery NEW

- Automated scheduled backups, manual on-demand export, restore drills, off-site/cloud backup replication, documented RTO/RPO targets.

5. Non-Functional Requirements

5.1 Performance

ID	Requirement
NFR-PRF.1	The system shall support at least 500 concurrent users for the MVP, with a scaling path to 5,000+ for the multi-branch phase.
NFR-PRF.2	Common page loads (dashboard, patient search, appointment booking) shall complete in under 2 seconds at the 95th percentile.
NFR-PRF.3	Standard API responses shall complete in under 500 ms at the 95th percentile under normal load.

5.2 Security

ID	Requirement
NFR-SEC.1	Authentication shall use JWT/OAuth2 with short-lived access tokens and refresh tokens; multi-factor authentication shall be available for admin and clinical roles. NEW
NFR-SEC.2	All access shall be governed by Role-Based Access Control with field- and record-level restrictions where clinically required.
NFR-SEC.3	Passwords shall be stored using a strong salted hash (e.g., bcrypt/argon2); plaintext passwords shall never be logged.
NFR-SEC.4	All network traffic shall be encrypted via TLS 1.2+; data at rest for sensitive fields (NID, diagnoses) shall be encrypted.
NFR-SEC.5	Every create/update/delete/view-of-sensitive-record action shall be captured in the tamper-evident audit log.
NFR-SEC.6	Automated daily backups with a minimum 30-day retention and periodic restore testing.
NFR-SEC.7	Session timeout after a configurable idle period, with automatic re-authentication for high-sensitivity screens. NEW

5.3 Availability & Reliability

- 99.9% uptime target for the core platform.
- Automated backups and a documented disaster recovery plan with defined Recovery Time Objective (RTO) and Recovery Point Objective (RPO). NEW
- Graceful degradation: front-desk token/queue functions should remain usable during brief backend interruptions via local caching. NEW

5.4 Usability & Accessibility NEW

- Bilingual UI (English/Bengali) selectable per user.
- Keyboard-first workflows for high-volume desks (registration, billing).
- WCAG 2.1 AA accessibility targets for the patient-facing portal.

5.5 Scalability & Maintainability

- Multi-hospital / multi-branch architecture (future-ready), microservice-compatible module boundaries.
- REST API with an optional GraphQL gateway; versioned APIs to avoid breaking integrations.
- Cloud deployment support (containerized via Docker, orchestration-ready).

5.6 Compliance NEW

- Medical record retention aligned with local health-ministry requirements.
- Architecture ready to support HIPAA/GDPR-equivalent controls for future international deployment.
- Support for government health-reporting data export formats (future phase).

6. System Architecture & Technology Recommendations

The following stack is a recommendation, not a hard constraint, and should be validated against the development team's expertise and hosting budget.

Layer	Recommendation
Frontend (Web)	React or Next.js, component library (e.g., Tailwind/shadcn), i18n framework for English/Bengali
Backend API	Node.js (NestJS/Express) or a similarly modular framework, REST + versioned endpoints
Database	PostgreSQL or MySQL as the primary relational store (patients, billing, HR); a document store (e.g., MongoDB) may be used for unstructured clinical attachments/logs
Authentication	OAuth2/JWT via an identity provider or self-hosted auth service
File/Media Storage	Object storage (S3-compatible) for scans, images, reports
Messaging/Notifications	SMS gateway, email service, WhatsApp Business API
Payments	Local mobile banking gateways (bKash, Nagad, Rocket) plus card processor
Deployment	Docker containers, CI/CD pipeline, staging + production environments
Monitoring	Centralized logging, uptime monitoring, error tracking (e.g., Sentry-equivalent)

> High-level architecture: a client layer (web + future mobile) talks to an API gateway which routes to modular services (Patient/EMR, Scheduling, Lab, Pharmacy, Billing, HRM, Reporting) sharing a common auth/RBAC service and a central relational database, with async messaging for notifications and background jobs (report generation, reminders, backups).

7. Data Model Overview (Key Entities)

This is not an exhaustive schema but identifies the core entities and their primary relationships to guide database design.

Entity	Key Relationships
Hospital / Branch	Has many Departments, Users, Patients (multi-branch ready)
User	Belongs to a Hospital; has a Role; may be linked to an Employee record and/or a Doctor profile
Patient	Has many Visits, Appointments, Prescriptions, Lab Orders, Bills, Admissions
Appointment	Belongs to a Patient and a Doctor; generates a Visit
Visit / Encounter	Belongs to a Patient; has Diagnoses, Clinical Notes, Prescriptions, Lab/Radiology Orders
Prescription	Belongs to a Visit; has many Prescription Items (medicine, dose, duration)
Lab Order / Test	Belongs to a Visit; has a Sample, Result(s), Approval status
Admission (IPD)	Belongs to a Patient; linked to a Bed, Ward, assigned Doctor/Nurses
Bill / Invoice	Belongs to a Patient/Visit/Admission; has many Line Items, Payments
Employee	Belongs to a Department; has Attendance, Leave, Payroll records
Inventory Item	Has Stock Batches (with expiry), linked to Purchase Orders and Suppliers
Audit Log Entry	References a User, an Action, and an affected Entity/Record ID

8. External Interface Requirements

8.1 User Interfaces

Responsive web application; role-specific dashboards; print-optimized layouts for prescriptions, lab reports, invoices, and discharge summaries.

8.2 Hardware Interfaces

- Barcode/QR scanners (sample labels, patient wristbands)
- Thermal receipt/label printers
- Biometric or RFID attendance devices
- Future: vital-sign monitor and lab analyzer interfaces (HL7/ASTM)

8.3 Software Interfaces

- SMS gateway API, Email (SMTP/API) service, WhatsApp Business API
- Mobile banking / card payment gateway APIs
- Future: national health information exchange (HL7/FHIR)

8.4 Communication Interfaces

HTTPS/REST for all client-server and third-party integrations; WebSocket channel recommended for live queue/token and bed-status updates.

9. Sample Use Cases

UC-01: Register a Walk-in Patient and Book an OPD Appointment

Actor	Receptionist
Precondition	Receptionist is logged in; patient is not yet in the system
Main Flow	1) Receptionist searches by phone/name to rule out an existing record. 2) Creates a new patient record with required fields. 3) Selects department/doctor and an available slot. 4) System generates a queue token and sends an SMS confirmation. 5) Receptionist collects the consultation fee.
Postcondition	Patient record and appointment exist; token is visible on the queue display.

UC-02: Doctor Consultation with Lab Order and E-Prescription

Actor	Doctor
Precondition	Patient is checked in and next in the doctor's queue
Main Flow	1) Doctor opens the patient's EMR and reviews history. 2) Records symptoms, diagnosis, and clinical notes. 3) Orders one or more lab tests. 4) Issues an e-prescription; system checks against recorded allergies. 5) Marks consultation complete, triggering billing for the consultation and ordered tests.
Postcondition	Visit record is complete; lab order appears in the Lab Technician's worklist; prescription is available to the Pharmacist and patient portal.

UC-03: In-Patient Admission Through Discharge

Actor	Doctor, Nurse, Accountant
Main Flow	1) Doctor recommends admission from OPD/ER. 2) Receptionist/Nurse allocates a ward and bed, generates an estimated cost. 3) Nurse records vitals and medication administration throughout the stay. 4) Doctor updates clinical notes and eventually issues discharge orders with a discharge summary. 5) Accountant finalizes the bill including room charges, procedures, medicines and lab tests; patient settles payment.
Postcondition	Bed is released and flagged for cleaning; discharge summary is available in the patient portal.

10. Release Plan — MVP and Future Phases

10.1 Phase 1 (MVP)

- Authentication & RBAC, System Configuration & Localization (bilingual)
- Dashboard, Patient Registration & Management
- Doctor & Department Management, Appointment & Queue Management
- OPD Consultation & EMR, Laboratory Management, Radiology (request/report upload)
- Pharmacy Management, Billing & Payments (cash, card, mobile banking)
- Admission, Ward & Bed Management
- HRM (Employee, Attendance, Leave, Payroll basics)
- CRM (Patient Feedback & Follow-up), Inventory Management
- Reports & Analytics, Notifications (SMS/Email), Audit Logs
- Blood Bank basic module, Consent capture **NEW – recommended for MVP**

10.2 Phase 2

- Telemedicine & Video Consultation
- Patient Mobile App, Doctor Mobile App
- OT full workflow (pre-op checklist, post-op recovery), Emergency Department triage workflow
- Ambulance dispatch with live status

10.3 Phase 3 (Future / Advanced)

- PACS/Radiology Image Viewer, Lab analyzer (HL7/ASTM) auto-import
- AI Clinical Decision Support, Voice-to-Text Clinical Notes
- e-Prescriptions with digital signatures at national-registry level
- IoT Integration (patient monitors, nurse call systems)
- Multi-branch & multi-tenant support, Business Intelligence dashboards
- HL7/FHIR interoperability, Government health reporting integrations
- Appointment chatbot / AI assistant, Predictive analytics for bed occupancy and staffing

11. Assumptions, Dependencies & Constraints

- Reliable internet connectivity is available at each hospital facility during operating hours.
- Staff will undergo structured training before go-live; a super-user/champion will be nominated per department.
- Existing paper or legacy digital records will be migrated in a defined data-migration project prior to cutover.
- Third-party SMS, mobile-banking, and email providers are available and contracted before Phase 1 launch.
- Hardware (barcode scanners, printers) is procured by the hospital per the specification provided by the development team.

12. Risks & Mitigations

Risk	Impact	Mitigation
Staff resistance to digital workflows	Slow adoption, reverting to paper	Phased rollout, hands-on training, in-app guided tours
Network downtime at facility	Inability to register/bill patients	Local caching for critical desk functions; offline queue token printing
Data breach of sensitive medical records	Legal and reputational damage	Encryption at rest/in transit, RBAC, regular security audits, MFA for privileged roles
Incomplete/incorrect data migration	Clinical errors from missing history	Migration dry-run, reconciliation reports, parallel-run period
Scope creep across 30 modules	Delayed MVP delivery	Strict phase gating as defined in Section 10; change-control process

13. Acceptance Criteria

The MVP shall be considered ready for go-live when:

- All Phase 1 modules listed in Section 10.1 pass functional QA against their FR-IDs with zero critical/high-severity open defects.
- Non-functional targets in Section 5 (page load, API latency, concurrent users) are verified under load testing.
- Role-based access has been penetration-tested for at least the Admin, Doctor, Receptionist, Pharmacist, and Accountant roles.
- A full backup-and-restore drill has been successfully executed.
- At least one end-to-end pilot (registration → consultation → lab → pharmacy → billing → discharge) has been run successfully in a staging environment with real hospital staff.

14. Appendix — Glossary

Term	Definition
Encounter / Visit	A single interaction between a patient and the hospital (OPD, IPD, or ER) that generates clinical and billing records
Token System	Sequential numbering used to manage patient queues at OPD or diagnostic counters
Critical Value	A lab result significantly outside the normal range requiring immediate clinical attention
RTO / RPO	Recovery Time Objective / Recovery Point Objective — disaster-recovery targets for downtime and data loss